

A Nonconvex Gauge with a Global Thy Angle

Candidate full affirmative answers to Problems 1 and 2 of arXiv:0902.2731

Autonomous proof candidate for human review

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Status. Candidate full affirmative answers, likely valid. An explicit positive-definite, absolutely homogeneous gauge on $\mathbb{R}^2$, smooth away from the origin, fails the triangle inequality, while its normalized polarization product satisfies the Cauchy–Schwarz–Bunjakowsky inequality. Its Thy angle also satisfies axiom (An 11). The only large algebraic step is accompanied by an exact rational Bernstein-basis certificate generated from the gauge itself.

1 The questions and the answer

For a positive-definite absolutely homogeneous gauge q , the source defines the polarization product, for nonzero x, y , by

$$\langle x \mid y \rangle_{\spadesuit} = \frac{1}{4}q(x)q(y) \left[q\left(\frac{x}{q(x)} + \frac{y}{q(y)}\right)^2 - q\left(\frac{x}{q(x)} - \frac{y}{q(y)}\right)^2 \right]. \quad (1)$$

Problem 1 asks whether such a q can fail the triangle inequality while $|\langle x \mid y \rangle_{\spadesuit}| \leq q(x)q(y)$ for all x, y . Problem 2 asks whether, in addition, for every independent x, y the map

$$t \longmapsto \angle_{\text{Thy}}(x, y + tx) = \arccos \frac{\langle x \mid y + tx \rangle_{\spadesuit}}{q(x)q(y + tx)}$$

is a decreasing homeomorphism from $\mathbb{R}$ onto $(0, \pi)$.

Theorem 1. *Both problems have affirmative answers. On $\mathbb{R}^2$, put $q(0) = 0$ and, in Euclidean polar coordinates,*

$$q(r \cos \theta, r \sin \theta) = r \left(1 + \frac{1}{10} \cos 4\theta \right). \quad (2)$$

Then q is continuous, positive definite, and absolutely homogeneous, but does not satisfy the triangle inequality. Nevertheless its product (1) satisfies CSB, and the associated Thy angle satisfies (An 11).

The rest of the packet proves the theorem. The conceptual point is that the small fourfold-symmetric inward dents are compatible with strict angular monotonicity of the polarization, even though corners of the kind excluded in the source are not.

$$= 1 + \delta \cdot (m_+ - m_-) + \frac{1}{4} \cdot \delta^2 \cdot K ,$$

with the real constant $K := (m_+ - m_-)^2 - \|2 \cdot \text{sign}(\vec{x}) + (m_+ + m_-) \cdot \text{sign}(\vec{y})\|^2$.

This calculation holds for all $\delta \in [0, \varepsilon]$. Hence, for a positive $\hat{\delta}$ that is almost 0, because of $m_+ - m_- > 0$ we have $P_\blacklozenge(\hat{\delta}) > 1$, hence, for the unit vectors $\hat{\delta} \cdot \text{sign}(\vec{x}) + (1 + \hat{\delta} \cdot m_+) \cdot \text{sign}(\vec{y})$ and $-\hat{\delta} \cdot \text{sign}(\vec{x}) + (1 - \hat{\delta} \cdot m_-) \cdot \text{sign}(\vec{y})$, the CSB inequality is not satisfied. $\square$

Corollary 3. For all $0 < r < 1$, the space $(\mathbb{R}^2, \|\cdot\|_{\text{Polygon}, r}, \langle \cdot, \cdot \rangle_\blacklozenge)$ does not fulfil the CSB inequality. Hence, there are vectors $\vec{x} \neq \vec{0} \neq \vec{y}$ such that the 'Thy angle' $\angle_{\text{Thy}}(\vec{x}, \vec{y})$ is not defined. Hence, for $0 < r < 1$, the space $(\mathbb{R}^2, \|\cdot\|_{\text{Polygon}, r}, \angle_{\text{Thy}})$ is not an 'angle space'.

As we mentioned in the beginning of the section, we formulate two open problems. We always deal with pseudonormed vector spaces $(X, \|\cdot\|)$ (that are **hw spaces** which are positive definite, that means $\|\vec{x}\| = 0$ only for $\vec{x} = \vec{0}$) with a non-convex unit ball.

Problem 1: Exists a pseudonormed vector space $(X, \|\cdot\|)$, such that $\|\cdot\|$ does not fulfil the triangle inequality, but $(X, \|\cdot\|, \langle \cdot, \cdot \rangle_\blacklozenge)$ satisfies the CSB inequality?

Problem 2: If Problem 1 is true, exists a pseudonormed vector space $(X, \|\cdot\|)$, such that $\|\cdot\|$ does not fulfil the triangle inequality, but $(X, \|\cdot\|, \langle \cdot, \cdot \rangle_\blacklozenge)$ satisfies the CSB inequality and the axiom (An 11) is fulfilled?

Good candidates for both questions are the *Hölder weights* $\|\cdot\|_p$ on $\mathbb{R}^2$, $\|(x, y)\|_p = \sqrt[p]{|x|^p + |y|^p}$, with $0 < p < 1$, but p almost 1.

7 A Generalization of the Thy Angle

Figure 1: Problems 1 and 2 in the source, PDF page 19.

2 Elementary properties and failure of convexity

Write

$$w(\theta) = 1 + \frac{1}{10} \cos 4\theta, \quad u_\theta = \frac{(\cos \theta, \sin \theta)}{w(\theta)}.$$

Since $9/10 \leq w \leq 11/10$, formula (2) gives a continuous, positive-definite, absolutely homogeneous gauge, and $q(u_\theta) = 1$. Away from the origin it also has the Cartesian formula

$$q(X, Y)^2 = \frac{(11X^4 + 14X^2Y^2 + 11Y^4)^2}{100(X^2 + Y^2)^3}. \quad (3)$$

Let $h = \pi/8$. Because $w(h) = w(-h) = 1$, the two vectors

$$u_+ = (\cos h, \sin h), \quad u_- = (\cos h, -\sin h)$$

have $q(u_+) = q(u_-) = 1$. Their midpoint is $(\cos h, 0)$, but

$$q\left(\frac{u_+ + u_-}{2}\right) = \frac{11}{10} \cos \frac{\pi}{8} > 1.$$

Thus the unit ball is not convex. More directly, the triangle inequality and homogeneity would force the left side to be at most $\frac{1}{2}(q(u_+) + q(u_-)) = 1$, a contradiction.

3 Reduction to strict angular monotonicity

For a base direction α and an oriented separation δ , define

$$C_\alpha(\delta) = \frac{1}{4} (q(u_\alpha + u_{\alpha+\delta})^2 - q(u_\alpha - u_{\alpha+\delta})^2). \quad (4)$$

This is exactly the normalized product $\langle u_\alpha \mid u_{\alpha+\delta} \rangle_\spadesuit$.

Lemma 1. *For every $\alpha \in \mathbb{R}$, the function C_α is strictly decreasing on $[0, \pi]$, with*

$$C_\alpha(0) = 1, \quad C_\alpha(\pi) = -1.$$

Before proving the lemma, note its two immediate consequences. Any pair of directions can, after exchanging the two vectors if necessary, be written with a separation in $[0, \pi]$. Lemma 1 then gives $|C_\alpha(\delta)| \leq 1$. Absolute homogeneity of (1) therefore gives CSB for all vectors. It remains only to establish the strict derivative assertion.

4 Exact positivity certificate

The gauge is invariant under rotation through $\pi/2$ and reflection in the first coordinate axis. For $0 < \delta \leq \pi/2$, rotation therefore reduces the derivative calculation to

$$0 \leq \alpha \leq \frac{\pi}{2}, \quad 0 < \delta \leq \frac{\pi}{2}.$$

Introduce rational half-angle coordinates

$$x = \tan \frac{\alpha}{2} \in [0, 1], \quad z = \tan \frac{\delta}{2} \in [0, 1].$$

For reference, a q -unit direction with half-angle parameter t is the rational vector

$$U(t) = \frac{10(1+t^2)^3}{11+12t^2+130t^4+12t^6+11t^8} (1-t^2, 2t). \quad (5)$$

Thus $u_\alpha = U(x)$ and, where the displayed fraction is finite,

$$u_{\alpha+\delta} = U\left(\frac{x+z}{1-xz}\right).$$

Equations (3) and (5) make $C_\alpha(\delta)$ a rational function of x, z .

The accompanying verifier performs the following calculation from scratch over $\mathbb{Q}[x, z]$. It forms the two rational vectors $U(x) \pm U((x+z)/(1-xz))$, evaluates (3), forms (4), differentiates, and cancels the polynomial gcd. The result has the form

$$-\frac{1}{\sin \delta} \frac{\partial C_\alpha}{\partial \delta} = \frac{(1+z^2)^4 P(x, z)}{H(x, z)}, \quad (6)$$

where P, H are primitive integer polynomials. (The primitive contents of the reduced numerator and denominator are exactly equal.) The identity first holds on the dense locus on which the half-angle expressions have no pole and then, after cancellation, by continuity throughout the relevant interior.

Here is the exact positivity certificate:

polynomial	bidegree	monomials	Bernstein coefficients	least coefficient
P	(128, 104)	6773	13545	12531744508246953
H	(128, 112)	7289	14577	1285453186679270785/28

Every tensor Bernstein coefficient on $[0, 1]^2$ is strictly positive. The least coefficient of P has tensor index $(0, 0)$; that of H has index $(1, 1)$. Since Bernstein basis functions are nonnegative and sum to one, this proves $P > 0$ and $H > 0$ on the whole square. Equation (6) consequently gives

$$\frac{\partial C_\alpha}{\partial \delta} < 0 \quad (0 < \delta \leq \pi/2). \quad (7)$$

The verifier uses Python’s exact **Fraction** arithmetic for the entire Bernstein conversion. It stores no precomputed P or H ; those polynomials are reconstructed from (2) on every run.

Completion of Lemma 1. It remains to transfer (7) to the second half-circle. Put $\varepsilon = \pi - \delta$ when $\pi/2 < \delta < \pi$. Antipodality, oddness of the normalized polarization in either argument, and reflection invariance give

$$\begin{aligned} C_\alpha(\pi - \varepsilon) &= -C(u_\alpha, u_{\alpha-\varepsilon}) \\ &= -C(u_{-\alpha}, u_{-\alpha+\varepsilon}) = -C_{-\alpha}(\varepsilon). \end{aligned}$$

Since $d\varepsilon/d\delta = -1$, differentiation yields

$$\frac{d}{d\delta} C_\alpha(\delta) = C'_{-\alpha}(\varepsilon) < 0$$

by (7). Finally, absolute homogeneity gives $u_{\alpha+\pi} = -u_\alpha$, so direct substitution into (4) gives the endpoint values 1 and -1 . $\square$

5 Axiom (An 11)

Let $a, b \in \mathbb{R}^2$ be linearly independent. Suppose first that b lies in the open counterclockwise half-plane from the ray a . The counterclockwise separation $\delta(t)$ from a to $b + ta$ is a decreasing homeomorphism

$$\delta : \mathbb{R} \longrightarrow (0, \pi), \quad \lim_{t \rightarrow -\infty} \delta(t) = \pi, \quad \lim_{t \rightarrow +\infty} \delta(t) = 0.$$

(After a Euclidean choice of direction coordinates with the ray a as the positive horizontal axis, this is simply the argument of $(s + t, r)$ for some $r > 0$.) Lemma 1 says that the normalized polarization cosine $C(\delta(t))$ is a strictly increasing homeomorphism from $\mathbb{R}$ onto $(-1, 1)$. Since arccos is a decreasing homeomorphism from $(-1, 1)$ onto $(0, \pi)$,

$$t \longmapsto \angle_{\text{Thy}}(a, b + ta)$$

is a decreasing homeomorphism from $\mathbb{R}$ onto $(0, \pi)$. If b lies in the clockwise half-plane, then $C(u_\alpha, u_{\alpha-\delta}) = C(u_{-\alpha}, u_{-\alpha+\delta})$ by reflection invariance, and the identical argument applies with base $-\alpha$. This is exactly (An 11), and completes the proof of Theorem 1.

6 Verification and scope

The exact verifier is `verify_certificate.py` in this packet. On the recorded run it reconstructed the rational Thy cosine with numerator and denominator bidegree $(96, 84)$, canceled a derivative gcd of bidegree $(64, 57)$, and certified all 28122 Bernstein coefficients described above. The nonconvexity witness and the symmetry reduction are independent of that calculation.

A bounded novelty search found the two questions in the 2009 source, the author’s later 2013 classification paper (which continues to discuss nonconvex balanced weights and concave-corner

obstructions), and later work using Thy angles in normed or n -normed spaces. It did not locate this gauge, this certificate, or a later solution of the two questions. Priority is not asserted; the result should be independently reviewed before citation.

References

- [1] V. W. Thürey, *Angles and Polar Coordinates In Real Normed Spaces*, arXiv:0902.2731 (2009).
- [2] V. W. Thürey, *Angles and a Classification of Normed Spaces*, *Annals of Functional Analysis* 4 (2013), no. 1, 114–137; arXiv:1207.0074.